

Practice: Modeling Real-World Problems

Problem 1: Menu Ranker

IPO

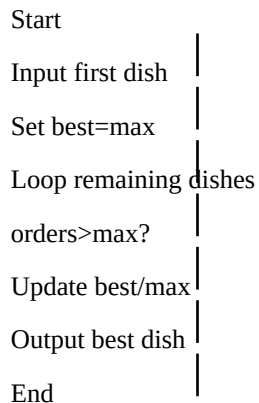
Input: Five dishes with order counts.

Process: Compare order counts, keep track of highest. If a tie occurs, keep the first dish encountered.

Output: Dish with highest orders.

Pseudocode

```
START
INPUT first dish and orders
SET bestDish = first dish
SET max = first orders
FOR remaining 4 dishes
    INPUT dish, orders
    IF orders > max THEN
        SET max = orders
        SET bestDish = dish
    ENDIF
ENDFOR
OUTPUT bestDish, max
END
```



Problem 2: Attendance Filter

IPO

Input: Student names and attendance percentages.

Process: Check each attendance. If attendance is below 75%, include student in reminder list.

Output: Reminder list.

Pseudocode

```
START
FOR each student
    INPUT name, attendance
    IF attendance < 75 THEN
        OUTPUT name
    ENDIF
ENDFOR
END
```

Start	
Input student	
attendance<75?	
Output name	
Next student	
End	

Special Conditions

Menu Ranker: If two dishes have the same highest orders, the first encountered dish is selected.
Attendance Filter: Students with exactly 75% attendance do NOT receive a reminder.